Lightweight Edge-Aware Mamba-Fusion Network for Weakly Supervised Salient Object Detection in Optical Remote Sensing Images

Gaojie Xing, Mengyin Wang, Fasheng Wang, Fuming Sun, and Haojie Li

***Abstract*— Despite the significant progress made in fully supervised salient object detection in optical remote sensing images (ORSI-SOD), these methods rely heavily on pixel-level annotations, which are time-consuming and labor-intensive. This situation has driven the development of weakly supervised ORSI-SOD methods. However, existing weakly supervised ORSI-SOD methods still face excessive model parameters and high computational complexity, hindering their flexibility and deployment in edge devices. To address these challenges, we propose the LightEMNet, a scribble-based, lightweight, and high-performance edge-aware network for ORSI-SOD. The network employs MobileNetV2 as its lightweight encoder backbone. To mitigate the suboptimal feature extraction performance caused by the lightweight architecture, we design a feature refinement layer (FRL) to refine the features extracted from the backbone, thereby generating guidance information while achieving better structural awareness and object localization. To realize better detail optimization, we introduce edge information extracted by a multiscale edge perception module (MEP) to regulate high-level features. Finally, considering the shortcomings of traditional convolution in global-awareness, we propose a Mamba-based cross-scale edge-semantic interaction (CESI) module to achieve efficient alignment of semantics and edges, which consequently enhances the representation consistency of the fused features and improves the model's adaptability to complex scenes. We verify the effectiveness of the LightEMNet through extensive experiments. The results demonstrate that the proposed LightEMNet exhibits competitive detection performance with only 4.81 M parameters. Codes and results are available at https://github.com/xingggao/LightEMNet**

***Index Terms*— Lightweight, Mamba, scribble, weakly supervised salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) aims to imitate human visual system and detect the most salient objects or regions in an image. As an important preprocessing step, SOD has promoted the development of multiple fields such as object tracking [1], [2] and video segmentation [3], [4]. SOD methods for single images of natural scenes [5], RGB and depth/thermal images [6], [7], and videos [8], [9] have been extensively studied in the community. In recent year, researchers start to focus on SOD in optical remote sensing images (ORSIs), named ORSI-SOD. ORSIs are generally obtained from high-altitude platforms like spacecraft and drones, and they are different from natural RGB images. As an emerging research topic, ORSI-SOD has attracted considerable attention from researchers.

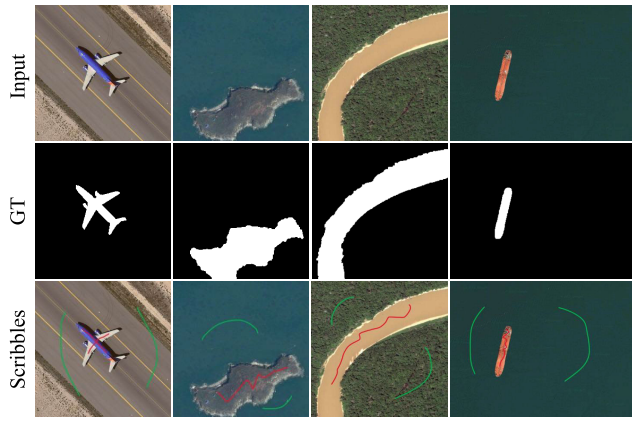


Fig. 1. Scribble and pixel-level label (GT).

Existing fully supervised ORSI-SOD methods rely on time-consuming manually labeled annotations, which limit their flexibility. Consequently, low-cost weakly supervised methods have emerged, including image-level labels [10], [11], point annotations [12], bounding boxes, and scribble annotations [13]. Among them, scribble annotations have gained more attention for their flexibility and efficiency in labeling objects. As shown in Fig. 1, compared to pixel-level annotations, scribble-based labeling offers a cost-effective alternative while retaining essential spatial information about foreground objects and background regions. However, this annotation efficiency comes at the expense of performance limitations, as the sparse nature of scribble annotations often leads to insufficient detail representation and compromised structural integrity of salient objects, manifesting as incomplete object structures and ambiguous boundaries. Consequently, the performance of weakly supervised ORSI-SOD methods continues

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to lag behind their fully supervised counterparts. Furthermore, many weakly supervised models are burdened with substantial computational overhead and excessive parameter requirements, presenting additional challenges for practical implementation.

Most of the ORSI-SOD models follow a local-to-local or global-to-local paradigm. The CNN-based encoder–decoder architecture demonstrates effective spatial information retrieval through its encoder, complemented by the decoder's progressive restoration of positional details. While this framework excels in local feature aggregation, it exhibits inherent limitations in capturing global contextual information. In contrast, the Transformer-based global-to-local paradigm leverages its robust contextual modeling capabilities to extract comprehensive global features, while its decoder effectively integrates local information for enhanced context awareness. Nevertheless, this paradigm is constrained by substantial parameter requirements and computational overhead, leading to excessively complex network architectures. A notable illustration is the ASNet [14], which employs a hybrid Transformer–CNN structure to synergize CNN's local feature extraction with Transformer's global perception capabilities, albeit at the cost of increased parameter complexity and computational demands.

In the ORSI-SOD domain, the incorporation of edge features has proved crucial for optimizing detection performance. Historically, two predominant approaches have emerged for edge information utilization. The first approach employs full-stage edge extraction, such as [15], where the authors utilize separate convolutional networks for simultaneous edge, semantic, and skeleton feature generation. While effective, this method typically results in computationally intensive and structurally complex networks. Alternatively, edge features can be constrained through shallow edge supervision, such as [16] and [17]. These approaches primarily achieve edge refinement without fully capitalizing on the potential of edge information to systematically guide the feature decoding process. Consequently, the effective integration of edge information necessitates a comprehensive framework that not only incorporates edge features but also strategically utilizes them to enhance the learning and representation of high-level semantic features throughout the network architecture. This requires careful consideration of how edge information can be optimally leveraged to establish a more robust detection network.

To address the aforementioned challenges, we propose a novel lightweight edge-aware weakly supervised framework that maintains competitive performance while achieving computational efficiency. Building upon the established understanding that backbone-extracted features can be categorized into low-level features (containing contour and texture information for detail refinement) and high-level features (possessing rich semantic and positional information), we implement a hierarchical optimization strategy for comprehensive feature exploration. Specifically, our framework incorporates three key components: a feature refinement layer (FRL), a multiscale edge perception (MEP) module, and a cross-scale edge-semantic interaction (CESI) module. The FRL, grounded in hierarchical optimization principles, mitigates information loss inherent in lightweight backbones while enhancing structural awareness and object localization precision. Recognizing the crucial role of deep-level features in edge construction, we introduce the MEP module to facilitate interaction between shallow and deep-level features, thereby improving structural focus and boundary localization capabilities. This module effectively bridges the gap between different feature layers through interactive utilization of low and high-level features for edge information extraction. Finally, although edge features provide local details, it is worth thinking about how to make full use of them to enhance the representation of salient objects. The Mamba model can utilize the edge to constrain object semantics, and semantic features can also promote edge refinement. This self-supervised method contributes to more precise feature reconstruction, resulting in more accurate and effective fusion of low and high-level features. Therefore, we propose CESI based on Mamba to empower global context modeling capability for edge-guided feature interaction. This innovative design achieves superior interaction results while maintaining low parameter complexity, ultimately enabling remarkable detection performance under scribble supervision conditions.

Our work makes the following distinct contributions.

1) We propose a scribble-based, lightweight, weakly supervised ORSI-SOD method. It offers dual advantages of parameter efficiency and cost-effective annotation utilization. This unique design has a significant benefit in resource-constrained application scenarios.

2) An FRL is designed to make up for the deficiency of a lightweight backbone in feature extraction. It adopts a hierarchical optimization strategy, from multiple views to further optimize the perception ability of the model.

3) A novel MEP module is designed to interactively utilize the features of different layers for edge exploration to achieve better edge extraction and background suppression. The module no longer follows the traditional bottom-up paradigm, but adopts the strategy of shallow layer extraction and high layer constraint. It realizes edge construction with the advantage of low parameters.

4) We propose a Mamba-based CESI module to make full use of edge information to guide contextual interaction, which takes advantage of Mamba in long sequence modeling, endows the ability of global perception to the model while maintaining low computational cost.

II. Related Work

A. Fully Supervised ORSI-SOD

With the development of deep learning, convolutional neural networks (CNNs) are widely used in object detection [18], [19], recognition [20], and SOD [21], [22]. It has been introduced into ORSI-SOD, leading to the emergence of CNN-based models. Li et al. [23] constructed the first dataset ORSSD and proposed an LVNet model for ORSI-SOD, which utilized a dual-stream pyramid and nested encoder–decoder structure. Subsequently, Zhang et al. [24] extended the dataset to establish EORSSD and proposed an end-to-end dense attention flow network. Later, to further refine the object boundary, many researchers introduced edge information as guidance.

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Zeng et al. [17] enhanced the object's boundary through constructing a difference perception mechanism in shallow layers. Gong et al. [15] proposed a two-stage network named ESGNet. The first stage generates edge, skeleton, and rough salient map, while another stage polishes the rough map, employing a coarse-to-fine strategy to achieve optimal results. With the application of the Transformer in computer vision, self-attention and multihead attention mechanisms have become popular in ORSI-SOD. Li et al. [25] utilized a self-attention mechanism to model the relationship between different-scale features for obtaining discriminative features. To simultaneously capture both local and global features, Yan et al. [14] proposed a dual-backbone network, which makes full use of the power of CNNs in local features extraction and Transformers in global context.

While these methods have demonstrated promising performance, they are inherently constrained by their reliance on large-scale, expensively annotated datasets. Furthermore, the incorporation of attention mechanisms significantly escalates both parameter quantities and computational overhead, rendering them particularly unsuitable for edge computing devices and consequently impeding their practical deployment in real-world applications.

B. Weakly Supervised ORSI-SOD

Considering the expensive labor costs of fully supervised methods, sparse annotation has become popular. Many researchers have proposed weakly supervised methods for SOD [5], [10], [26], [27], [28]. Currently, sparse annotations have been extended to several downstream tasks, including ORSI-SOD. Initially, Huang et al. [29] first employed scribble labels as supervision and used a boundary generation network to improve localization problems. Ma et al. [13] utilized a scribble embedding network and a multibranch architecture to enhance the discriminative ability for foreground. Li et al. [30] established a constraint relationship between the local model and enhanced model through a salient region correcting mechanism to automatically calibrate salient regions. Yan et al. [52] proposed a one-stop multiscale reconciliation attention network, named OMNet, which gradually recovers object structures by processing multilevel features. Moreover, dual-stream encoding structures are also introduced into the field of weakly supervised ORSI-SOD. The GLSIN proposed by Yan et al. [31] used a CNN–Transformer encoding structure and explored the effective fusion of local details from the CNN and global features from the Transformer. Scribble annotation is also widely used in other areas of remote sensing, including building extraction, road segmentation. For instance, Chen et al. [32] leveraged edge information to reconstruct building structures from scribble labels, while precisely constraining object boundaries through an edge-aware loss function.

These findings provide valuable insights for weakly supervised ORSI-SOD. However, due to the inherent limitations of sparse annotations, a noticeable performance gap persists when compared to fully supervised approaches. Critical challenges, including suboptimal object boundary refinement, inadequate foreground–background differentiation, and computational inefficiency due to excessive parameters, remain unresolved. To address these limitations, we propose a novel weakly supervised framework that achieves superior detection accuracy while maintaining remarkable parameter efficiency.

C. Mamba in Computer Vision

Recent advancement in state-space models (SSMs) [33] has shown remarkable potential for long-sequence modeling, offering a new paradigm for ORSI-SOD. In contrast to Transformer-based architectures, SSMs offer superior computational efficiency by maintaining linear complexity with respect to sequence length, while simultaneously preserving robust long-range dependency modeling capabilities. Many studies have shown that Mamba also has outstanding performance in vision tasks.

Zhan et al. [34] proposed a dual Mamba-driven RGB-D fusion network, named MambaSOD, which employs dual-branch visual Mamba blocks as a backbone for feature extraction. The relationship between multimodal features is captured through the cross-modal fusion-Mamba module. Mamba has also gained widespread attention in medical image segmentation, and several Mamba-based models have achieved great results. The U-Mamba [35] shows better performance than traditional CNN- and Transformer-based methods. Xing et al. [36] proposed SegMamba, which enhances the capability to process high-dimensional features in 3-D medical images through an improved multidirectional scanning mechanism.

Currently, Mamba is also successfully applied to remote sensing. For instance, MiM-ISTD [37] adopts a Mamba-in-Mamba architecture, employing outer and inner Mamba blocks to achieve effective infrared small-target detection. RS3Mamba [38] utilizes a visual state-space model to construct an auxiliary branch, providing additional global information to the convolution-based main branch. To the best of our knowledge, Mamba has not been employed in ORSI-SOD. Inspired by Mamba's strong performance in vision tasks, we leverage its strengths to improve the fusion performance of edge details and semantics.

III. Proposed Method

A. Overall Architecture

Fig. 2 illustrates the structure of our proposed LightEMNet, which consists of four components: an encoder layer, FRL, MEP, and CESI. To capture salient objects with complex shapes and varying scales from ORSIs, the encoder must extract rich multilevel and multiscale features. As we aim to build a lightweight model, we use MobileNet-V2 as the backbone with the first 17 residual blocks by deleting the last three layers. The backbone is divided into five blocks, denoted as E^t $(t = 1, 2, 3, 4, 5)$, the output can be described as $f_e^i \in \mathbb{R}^{c_i \times h_i \times w_i}$, where h_i and w_i are $(352/2^t)$, and $c_i \in \{16, 24, 32, 96, 320\}$. To preserve more spatial information, we apply a $2\times$ upsampling operation to the resolution of each layer. The FRL is designed to compensate for the coarse content caused by the lightweight backbone. It consists of

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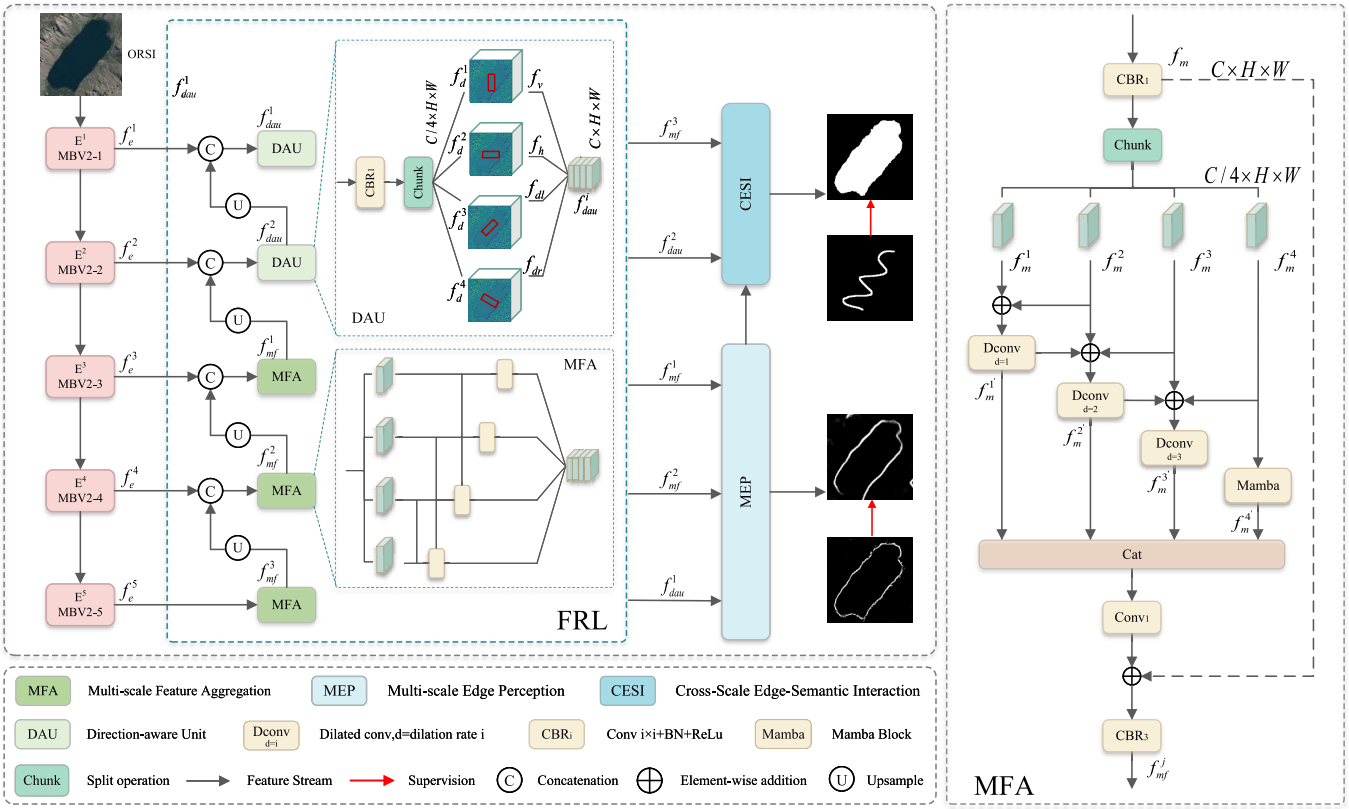


Fig. 2. Overall architecture of our proposed LightEMNet.

two parts: the shallow layers aim to improve the perception of object orientation through a direction-aware unit (DAU), and the high layers aim to enhance the localization ability through a multibranch structure. The MEP module refines the ability to suppress background interference based on the spatial attention and channel attention, and realizes accurate edge extraction through the strategy of shallow-layer extraction and high-layer constraint. Finally, the CESI module leverages Mamba's power in long-range modeling to guide the interaction of edge and high-level features, achieving better feature fusion and segmentation results.

B. Feature Refinement Layer

As shown in Fig. 2, the FRL consists of two units: DAU and MFA. It is widely acknowledged that the pursuit of lightweight model architectures typically entails a fundamental tradeoff with compromised feature extraction capacity. Furthermore, conventional pretrained backbones often demonstrate limited adaptability to the unique characteristics of ORSIs, rendering feature enhancement strategies particularly critical. To overcome these inherent challenges, we introduce the FRL, which employs a top-down feature propagation mechanism and hierarchical optimization strategy to achieve more effective feature refinement.

1) Direction-Aware Unit: In ORSIs, salient objects frequently exhibit diverse orientations and complex topological characteristics, posing significant challenges for the extraction of comprehensive structural details. Traditional symmetric convolution operations demonstrate limited capability when processing objects with specific directional attributes. To overcome this limitation, drawing inspiration from [39], we leverage asymmetric convolution at the low-level feature extraction stage, as shown in Fig. 2. This strategy enhances the model's capacity to discern directional patterns and structural information while simultaneously reducing redundant features. Specifically, our DAU incorporates four distinct asymmetric convolution kernels, enabling the model to effectively capture object orientations along four primary axes: horizontal, vertical, and two diagonal directions. We first get the fused features by concatenating the current feature with the feature from the previous layer, and then perform convolution and chunk operation to obtain the four feature blocks f_d^i $(i = 1, 2, 3, 4)$

$$f_c = \text{Concat}\left(f_e^i, U\left(f_{\text{dau}}^2\right)\right), \quad i = 1, 2 \tag{1}$$

$$f_d^i = \text{Chunk}(\text{CBR}(f_c)), \quad i = 1, 2, 3, 4 \tag{2}$$

where CBR() is a series of operations: Conv, BN, RELU, and $U()$ is the upsampling operation. When $i = 2$, replace f_{dau} with f_{mf}^1 in (1). Chunk() is the split operation. After that, we conduct different direction-aware operations on each block to obtain four directional features, f_h, f_v, f_{dl}, and f_{dr}. This multidirectional learning strategy enables comprehensive analysis of the shape, boundary, and textural details of objects, avoiding information loss caused by single-directional perception. After that, we perform a feature aggregation operation to fuse the four features. This multidirectional perception and aggregation mechanism significantly enhances the model's

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adaptability to object boundaries and complex topological structure. The process can be shown as follows:

$$\begin{cases} f_v = \mathrm{Conv}_v\left(f_d^1\right) \\ f_h = \mathrm{Conv}_h\left(f_d^2\right) \\ f_{dl} = \mathrm{Conv}_{dl}\left(f_d^3\right) \\ f_{dr} = \mathrm{Conv}_{dr}\left(f_d^4\right) \end{cases} \tag{3}$$

$$f_{\mathrm{dau}} = \mathrm{Concat}(f_h, f_v, f_{dl}, f_{dr}) \tag{4}$$

where $\mathrm{Conv}_v, \mathrm{Conv}_h, \mathrm{Conv}_{dl}, \mathrm{Conv}_{dr}$ denote the asymmetric convolution of 1×5, 5×1, and two diagonal convolution kernels, respectively. f_{dau} is the output of the DAU.

2) Multiscale Feature Aggregation: The direction and contour information of a salient object tends to diminish in high-level features. Therefore, the effect of applying DAU to high-level features is not obvious. In contrast, semantic and location information is rich in high-level features, which are more critical for recognizing and localizing salient objects. We design a multiscale feature aggregation (MFA) module which contains a cross-scale feature propagation process, as shown in Fig. 2. It is well known that large kernels increase the receptive field but also increase computational cost, while small kernels reduce the range of perception but weaken their spatial modeling capability. Consequently, to balance the disparity in performance caused by the appearance difference between large-scale and small-scale objects, we introduce a Mamba branch. The MFA module employs a parallel tri-branch architecture to gradually expand the receptive fields while utilizing a Mamba branch to compensate for the shortcomings of the convolution operation in global perception. Specifically, we first perform a concatenation operation between the current-level features and the output of the previous MFA and then adjust the channels through 1×1 convolution to maintain consistency in dimension. Finally, through the chunk operation, four feature blocks are obtained. The whole propagation process is described as follows:

$$\begin{cases} \mathrm{CBR}\left(\mathrm{Concat}\left(f_e^i, U\left(f_{mf}^{i+1}\right)\right)\right), & i = 3, 4 \\ \mathrm{CBR}\left(f_e^i\right), & i = 5 \end{cases} \tag{5}$$

$$f_m^i = \mathrm{Chunk}(f_m), \quad i = 1, 2, 3, 4. \tag{6}$$

Then, cross-scale interactive learning is performed through the following process:

$$f_m^{i'} = \begin{cases} \mathrm{DConv}_{d=i}\left(f_m^i \oplus f_m^{i+1}\right), & i = 1 \\ \mathrm{DConv}_{d=i}\left(f_m^{(i-1)'} \oplus f_m^i \oplus f_m^{i+1}\right), & i = 2, 3 \\ \mathrm{Mamba}\left(f_m^i\right), & i = 4 \end{cases} \tag{7}$$

where $\mathrm{DConv}_{d=i}()$ is the dilated convolution with a dilation rate of i, Mamba() represents the Mamba block, and $\oplus$ is the element-wise addition. Subsequently, convolution and residual connection are applied to the concatenated result f_{mm} to produce the output f_{mf}^i $(i = 1, 2, 3)$

$$f_{mm} = \mathrm{Concat}\left(f_m^{1'}, f_m^{2'}, f_m^{3'}, f_m^{4'}\right) \tag{8}$$

$$f_{mf}^i = \mathrm{CBR}(\mathrm{Conv}_{1\times1}(f_{mm}) \oplus f_m), \quad i = 1, 2, 3. \tag{9}$$

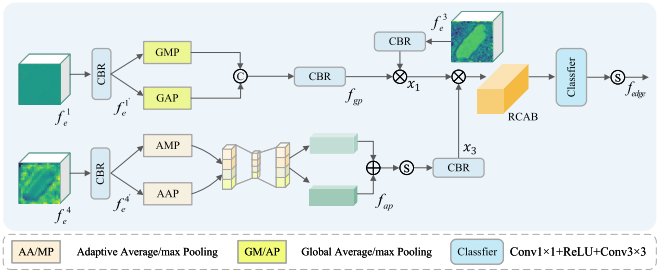


Fig. 3. Details of the MEP module.

C. MEP Module

In SOD tasks, edge features play a crucial role in guiding the generation of salient maps. To leverage edge information, existing ORSI-SOD models rely on an additional network to extract edge features, which often entails abundant computational costs and parameters. Alternatively, some methods impose constraints on edge features through shallow edge supervision, which merely achieve edge sharpening without making full use of edge details. To address these issues, we propose the MEP module. Guided by the strategy of shallow-level extraction and high-level constraint, the process involves performing edge extraction at shallow layers and leveraging semantic information for weighted optimization. As illustrated in Fig. 3, the inputs of MEP are the features f_e^1, f_e^3, and f_e^4 extracted by the backbone. First, we apply 1×1 convolutions to learn internal representation $f_e^{i'}(i = 1, 3, 4)$. Then, adaptive max-pooling and average pooling operations are applied to $f_e^{1'}$ to get the enhanced feature f_{en}. Simultaneously, we use $f_e^{3'}$ to perform the first weight optimization on f_{gp} to obtain x_1, which can be expressed as

$$f_e^{i'} = \mathrm{CBR}\left(f_e^i\right), \quad i = 1, 3, 4 \tag{10}$$

$$f_{gp} = \mathrm{CBR}\left(\mathrm{Concat}\left(\mathrm{GMP}\left(f_e^{1'}\right), \mathrm{GAP}\left(f_e^{1'}\right)\right)\right) \tag{11}$$

$$x_1 = f_{gp} \otimes f_e^{3'} \tag{12}$$

where GMP() and GAP() denote the global max-pooling and average pooling, respectively, and $\otimes$ denotes element-wise multiplication. Then, $f_e^{4'}$ is enhanced through a pooling operation and an MLP to improve channel interaction and through convolution to obtain x_3. Furthermore, x_3 is utilized to further strengthen x_1. Finally, a classifier and channel residual attention are employed to segment the single-channel edge map f_{edge}. The above process can be represented as follows:

$$f_{ap} = \mathrm{MLP}\left(\mathrm{AMP}\left(f_e^{4'}\right)\right) \oplus \mathrm{MLP}\left(\mathrm{AAP}\left(f_e^{4'}\right)\right) \tag{13}$$

$$x_3 = \mathrm{CBR}\left(\mathrm{Sigmoid}\left(f_{ap}\right)\right) \tag{14}$$

$$f_{\mathrm{edge}} = \mathrm{Sigmoid}(\mathrm{Classifier}(\mathrm{RCAB}(x_1 \otimes x_3))) \tag{15}$$

where AMP() and AAP() are adaptive max-pooling and average pooling, respectively, MLP() is a multilayer perceptron, RCAB() is the channel residual attention module, and Classifier() denotes the edge classifier consisting of Conv() and ReLu(). Sigmoid() is the activation function. This process significantly enhances the model's edge segmentation performance.

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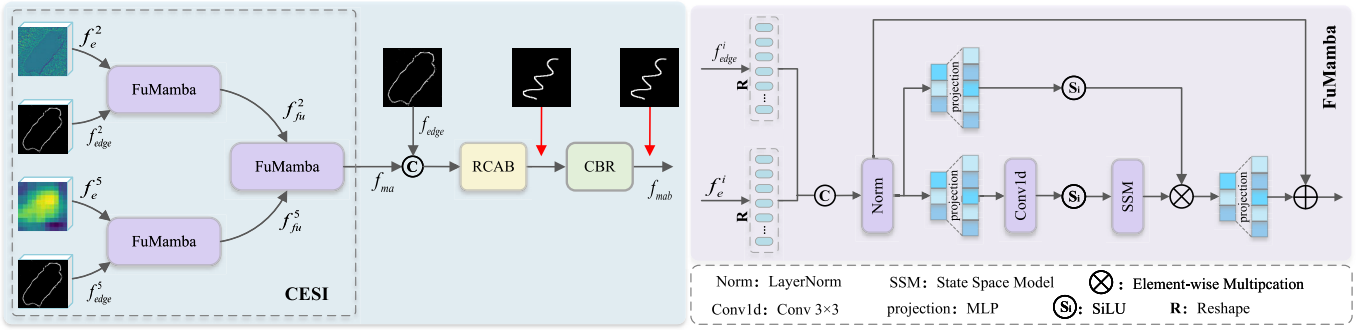


Fig. 4. Structure of the CESI module.

D. CESI Module

Feature fusion operations in SOD models predominantly employ self-attention or multihead attention mechanisms, which have become prevalent choices due to their exceptional capacity for capturing global contextual dependencies. However, these approaches are inherently constrained by substantial computational overhead and memory requirements, posing significant challenges for resource-constrained applications. Moreover, traditional convolution operations exhibit low costs and tend to overlook long-range dependency. To better balance these two situations, we propose the CESI module, as shown in Fig. 4. It introduces the highly efficient Mamba mechanism for feature fusion. To prevent the high computational costs caused by cascaded fusion, the inputs of the CESI module only use representative shallow-layer f_e^2 and high-layer f_e^5. And they, respectively, fuse with f_{edge} and obtain the typical shallow-level and high-level fusion features. First, we perform a convolution and upsampling operation on f_{edge} to ensure that it has the same spatial dimensions as f_e^2 and f_e^5, respectively. These features are then flattened into 1-D sequences and concatenated before sending to the Mamba blocks. The process is described as follows:

$$f_{\text{edge}}^i = U\left(\text{Conv}_{3\times3}\left(f_{\text{edge}}\right)\right), \quad i = 2, 5 \tag{16}$$

$$f_c^i = \text{Concat}\left(\text{Reshape}\left(f_e^i\right), \text{Reshape}\left(f_{\text{edge}}^i\right)\right), \quad i = 2, 5 \tag{17}$$

$$f_{fu}^i = \text{Mamba}\left(f_c^i\right), \quad i = 2, 5. \tag{18}$$

Subsequently, we perform secondary fusion on the fused features and generate CESI's output f_{ma}

$$f_{tc} = \text{Concat}\left(\text{Reshape}\left(f_{fu}^2\right), \quad \text{Reshape}\left(f_{fu}^5\right)\right) \tag{19}$$

$$f_{\text{ma}} = \text{Mamba}(f_{tc}) \tag{20}$$

where Reshape() is the reconstruction operation. Subsequently, the combined feature f_{ma} is passed through the decoder to produce the final output. The decoding process firstly performs a concatenation operation with f_{edge}, followed by RCAB and CBR operations, and finally generates the salient map f_{mab}:

$$f_{\text{mab}} = \text{CBR}\left(\text{RCAB}\left(\text{Concat}\left(f_{\text{ma}}, f_{\text{edge}}\right)\right)\right) \tag{21}$$

E. Loss Function

As shown in Fig. 2, our loss function consists of two parts: the edge loss and the saliency loss. To obtain clear boundary information, we use the binary cross-entropy loss L_{BCE} to constrain the edge feature map generated by MEP. The L_{BCE} loss is calculated as follows:

$$\mathcal{L}_{\text{BCE}} = \sum_j \left[\text{Edge} \log ep + (1 - \text{Edge}) \log(1 - ep)\right] \tag{22}$$

where j denotes the pixel index, Edge is the precomputed edge map, and ep is the edge prediction map. So, our edge loss is defined as

$$\mathcal{L}_e = \mathcal{L}_{\text{BCE}}\left(\text{Edge}, S^e\right) \tag{23}$$

where S^e donates our edge prediction map f_{edge} generated by MEP. As illustrated in Fig. 4, the saliency loss is also divided into two components. Due to the presence of unknown category pixels in the scribble annotations, we employ the partial cross-entropy loss L_{CE} and gated structure-aware loss L_{Gate} [26] to constrain coarse saliency map S^c and refined map S^r. L_{CE} loss is calculated as follows:

$$\mathcal{L}_{CE} = \sum_{p \in j} u_p \log S_p(p) \tag{24}$$

where u_p indicates whether the pixel is labeled, labeled as 1, and 0 otherwise. S_p donates the probability of the sample p being predicted as true. Our saliency loss is defined as

$$\begin{aligned} L_s = {} & L_{CE}\left(S^c, gt\right) + \alpha \cdot L_{\text{Gate}}\left(S^c, gt\right) \\ & + L_{CE}\left(S^r, gt\right) + \alpha \cdot L_{\text{Gate}}\left(S^r, gt\right) \end{aligned} \tag{25}$$

where gt donates the scribble annotations, and we set α to 0.3. Our final loss function can be defined as follows:

$$L_{\text{total}} = \mathcal{L}_e + \mathcal{L}_s. \tag{26}$$

The hybrid loss enables the scribble annotations to propagate to the foreground and realizes effective edge alignment.

IV. Experiments

A. Experimental Setup

1) Datasets: To verify the effectiveness of our method, we conduct extensive experiments on the ORSSD and EORSSD datasets. ORSSD [23] is the first dataset

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TABLE I
EXPERIMENTAL RESULTS OF THE COMPARISONS BETWEEN THE LIGHTEMNET AND SOTA ORSI-SOD METHODS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Methods	Params(M)	GFLOPs	EORSSD						ORSSD					
			F_β^{max}	F_β^{mean}	E_ξ^{max}	E_ξ^{mean}	S_α	$\mathcal{M}$	F_β^{max}	F_β^{mean}	E_ξ^{max}	E_ξ^{mean}	S_α	$\mathcal{M}$
Fully supervised ORSI-SOD														
EMFINet[22]	107.26	480.9	.8720	.8486	.9711	.9604	.9290	.0084	.9002	.8856	.9737	.9671	.9366	.0109
MJRBM[22]	43.54	95.7	.8656	.8239	.9646	.9350	.9197	.0099	.8842	.8566	.9623	.9415	.9204	.0163
CoorNet[22]	4.09	21.1	.8778	.8620	.9696	.9646	.9289	.0083	.9129	.9002	.9790	.9746	.9380	.0098
SeaNet[23]	2.76	1.7	.8649	.8519	.9710	.9651	.9208	.0073	.8942	.9772	.9767	.9722	.9260	.0105
ERPNet[23]	56.48	87.2	.8632	.8304	.7554	.9603	.9210	.0089	.8974	.8745	.9710	.9566	.9254	.0135
ACCoNet[23]	102.55	180.0	.8837	.8552	.9727	.9653	.9290	.0074	.9149	.8971	.9796	.9754	.9437	.0088
GeleNet[23]	25.45	11.7	.8923	.8781	.9828	.9766	.9376	.0064	.9254	.9128	.9860	.9815	.9469	.0079
SFANet[24]	25.1	7.7	.8833	.8680	.9769	.9726	.9349	.0058	.9192	.9063	.9830	.9789	.9453	.0077
ASNet[24]	65.43	20.1	.8959	.8770	.9783	.9745	.9345	.0055	.9172	.9072	.9803	.9764	.9441	.0081
MEANet[24]	3.3	11.3	.8766	.8678	.9658	.9658	.9282	.0070	.9033	.8934	.9730	.9730	.9340	.0098
SOLNet[24]	6.5	8.1	.8609	.8513	.9577	.9577	.9171	.0078	.9012	.8946	.9689	.9689	.9284	.0111
Weakly supervised/Unsupervised SOD														
WSSA[20]	15.53	70.6	.778	.771	.925	.901	.852	.015	.814	.806	.908	.908	.848	.027
SCWS[21]	63.54	53.9	.721	.626	.836	.737	.654	.094	.801	.722	.886	.799	.725	.096
WLSNet[22]	162.38	105.4	.427	.413	.696	.677	.678	.046	.514	.505	.729	.720	.696	.075
SSFaM[24]	312.85	1321.6	.724	.706	.880	.864	.831	.040	.753	.740	.899	.883	.836	.051
SBANet[22]	21.2	371.9	.816	.798	.938	.922	.861	.014	.837	.825	.939	.919	.863	.026
SHNet[22]	20.71	110.5	.784	.761	.923	.915	.868	.018	.791	.779	.915	.909	.856	.034
PAFR[23]	60.77	222.7	.826	.796	.949	.921	.893	.012	.843	.827	.946	.926	.893	.021
OMNet[24]	82.5	104.7	.852	.835	.954	.935	.906	.009	.867	.859	.958	.939	.902	.015
Ours	4.81	23.03	.847	.835	.962	.944	.887	.008	.884	.875	.969	.950	.900	.012

for ORSI-SOD covering plentiful scene types, of which 600 images are used for training and the remaining for testing. EORSSD [24] dataset is an extension of the ORSSD, adding images with semantic complexity and more challenging attributes. This dataset comprises 2000 ORSIs, each accompanied by pixel-level ground truth. Among them, 1400 images form the training set and 600 images form the testing set. Moreover, due to the random nature of manual scribble, we use the annotations provided by Li et al. [11] to ensure the fairness of our experiments.

2) Implementation Details: Our proposed model is implemented on PyTorch with an NVIDIA GeForce RTX 4090 GPU. We resize the input image and scribble annotations to 352 × 352. Additionally, the datasets are augmented fourfold to enhance the diversity and robustness of the training data. Our LightEMNet is trained using Adam optimizer for 60 epochs with a batch size of 6 and base learning rate of $1e^{-4}$, which will decay to $1e^{-5}$ after 40 epochs.

3) Evaluation Metrics: We evaluate the performance of our proposed method by quantitatively comparing it with other models using various evaluation metrics, including E-measure (E_ξ) [40], F-measure (F_β) [41], S-measure (S_α) [42], mean absolute error (MAE), precision–recall (PR) curve, and F-measure curve.

B. Comparison With SOTAs

We conduct comparative experiments between the proposed LightEMNet and 19 state-of-the-art (SOTA) competing methods. Through comprehensive qualitative and quantitative analyses, the advantages of LightEMNet are clearly demonstrated.

The selected 19 competing methods contain 11 fully supervised ORSI-SOD methods: EMFINet [43], MJRBM [44], CoorNet [45], SeaNet [46], ERPNet [47], ACCoNet [48], GeleNet [17], SFANet [32], ASNet [17], MEANet [49], SOLNet [50], and 8 weakly supervised methods: WSSA [26], SCWS [51], WLSNet [25], SSFaM [14], SBANet [29], SHNet [13], PAFR [36], and OMNet [52]. To ensure fair comparison, the results of the competing methods are obtained by running the released codes or directly using the results in the literature. For weakly supervised NSI-SOD methods, we retrain and test them using the same datasets and settings.

1) Quantitative Comparison: We compare the LightEMNet with the above 19 methods on EORSSD and ORSSD using 6 evaluation metrics ($F_\beta^{\max}$, F_β^{mean}, $E_\xi^{\max}$, E_ξ^{mean}, S_α, and MAE). The quantitative results are shown in Table I. We can see that, due to the limitations of scribble labels, there is a certain gap between weakly supervised methods and fully supervised methods, including SFANet [32] and ASNet [17]. Nonetheless, our LightEMANet demonstrates superior performance compared to several fully supervised ORSI-SOD methods, for example, CorrNet [45], ERPNet [47], and the lightweight SOLNet [50].

The results on the EORSSD demonstrate that our method achieves outstanding performance compared to other weakly supervised methods. When compared to the SOTA method,

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OMNet [52], our approach achieves significant performance improvements in several key metrics. Specifically, the LightEMNet shows improvements of 0.8%, 0.9%, and 0.1% in $E_\xi^{\max}$, E_ξ^{mean}, and MAE, respectively. It is worth noting that our LightEMNet has further narrowed the performance gap between fully supervised and weakly supervised methods, especially in MAE. The MAE gap between LightEMNet (0.8%) and the best supervised method, ASNet (0.55%), is reduced to 2.5%. This progress demonstrates that our proposed method can achieve comparable performance to fully supervised methods without expensive pixel-level supervision.

On the ORSSD dataset, our LightEMNet achieves significant improvements compared to OMNet [52]. Specifically, it achieves 1.8% improvement in the $F_\beta^{\max}$, 1.9% in F_β^{mean}, 1.1% in $E_\xi^{\max}$, 1.1% in E_ξ^{mean}, and 0.3% in MAE. Although there is a slight gap compared to OMNet in S_α, our superiority lies in significantly reducing the overall parameters and computational cost. The Params of our method are 16 times smaller than those of the OMNet, allowing us to maintain lightweight while achieving comparable performance. Meanwhile, the MAE gap between the LightEMNet and the best fully supervised method, SFANet, has been narrowed (0.73% $\rightarrow$ 0.43%), realizing significant progress in weakly supervised performance.

Fig. 5 illustrates the PR curves and F-measure curves on the EORSSD and ORSSD datasets. Compared to weakly supervised models (e.g., PAFR, SBANet, and SSFam), the PR curve of the LightEMNet (the red curve) is closer to the upper right corner both in EORSSD and ORSSD, which demonstrates the effectiveness of our weakly supervised method. However, our method still lags slightly behind some latest fully supervised methods. Meanwhile, the performance of our method (the dashed red curve) also surpasses existing weakly supervised methods on the F-measure curve. Notably, the F-measure curve of the LightEMNet exhibits an intertwined trend with fully supervised methods, indicating that our method has significantly narrowed the gap with fully supervised methods.

2) Qualitative Analysis: Fig. 6 illustrates the qualitative comparison results. These scenarios include complex backgrounds, multiscale targets, shadow interference, and intricate structure. Our LightEMNet shows excellent performance. For instance, the first four rows demonstrate the ability to detect objects with complex backgrounds and shadow interference, showing that the LightEMNet is more capable of handling disturbance. In the fifth and sixth rows, the images have lower contrast, making the detection in areas with similar colors particularly difficult. It can be observed that our model can overcome this difficulty. The seventh and eighth rows contain multiple objects, and our model can recognize these objects more accurately, reducing the possibility of missed detection. The ninth and tenth rows illustrate the model's capability to detect objects with intricate topological structure. It is observed that our LightEMNet can segment the details of the target areas more completely.

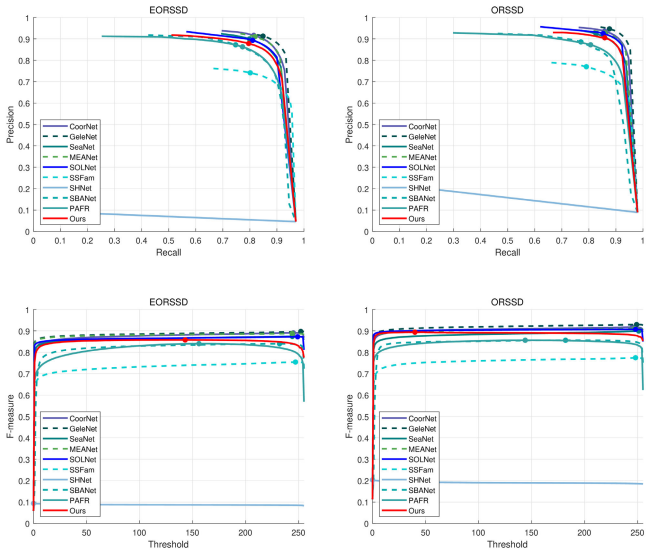


Fig. 5. PR and F-measure curves on the two datasets.

C. Ablation Studies

1) Ablation Experiments for Different Components in the LightEMNet: To verify the significance of each module in the model, we conduct the following experiments: a) baseline, b) baseline + FRL, c) baseline+MEP, d) baseline+MEP+CESI, and e) LightEMNet. We use MobileNet-V2 to extract multiscale features and fuse the features in an interactive manner for ORSI-SOD. The results are shown in Table II. It is clear that compared to the baseline a), by sequentially adding FRL b), MEP c), and CESI d) to the baseline, the overall performance is gradually improved, and the LightEMNet e) gets the best performance. We can see that adding any one of the three modules, the performance shows a certain degree of increase on most of the metrics. Meanwhile, the additional parameters per module are limited to 0.17–1.96 MB and maintaining a lightweight nature. Such performance improvement further verifies the importance of each module in our proposed LightEMNet.

Meanwhile, to intuitively demonstrate the role of individual modules, we also visualized the intermediate features generated from each module in various scenarios. We, respectively, selected the results from the first DAU and the second MFA in the FRL for presentation, as illustrated in Fig. 7(d) and (f). Compared to (c) and (e), the results of (d) and (f) show that the addition of DAU significantly enhances the shallow-layer features, and MFA also improves the localization capability of high-level features. The results of (g) highlight the effectiveness of MEP in edge extraction. The results of LightEMNet (h) indicate that CESI can effectively integrate multiple features, enhancing detection performance while maintaining the object's completeness.

2) Effectiveness of the FRL Module: As shown in Fig. 2, FRL consists of DAU and MFA. In this subsection, we verify the effectiveness of FRL, a) all_DAU, b) all_MFA, c) w/o Mamba, and d) Ours. As shown in Table III, most evaluation metrics shows obvious decrease on a) and b), which demonstrates that FRL with only DAU or MFA components cannot fully adapt to multilevel features, further validating the effectiveness of the hierarchical optimization strategy in FRL. Experiment c) replaces the Mamba branch in MFA

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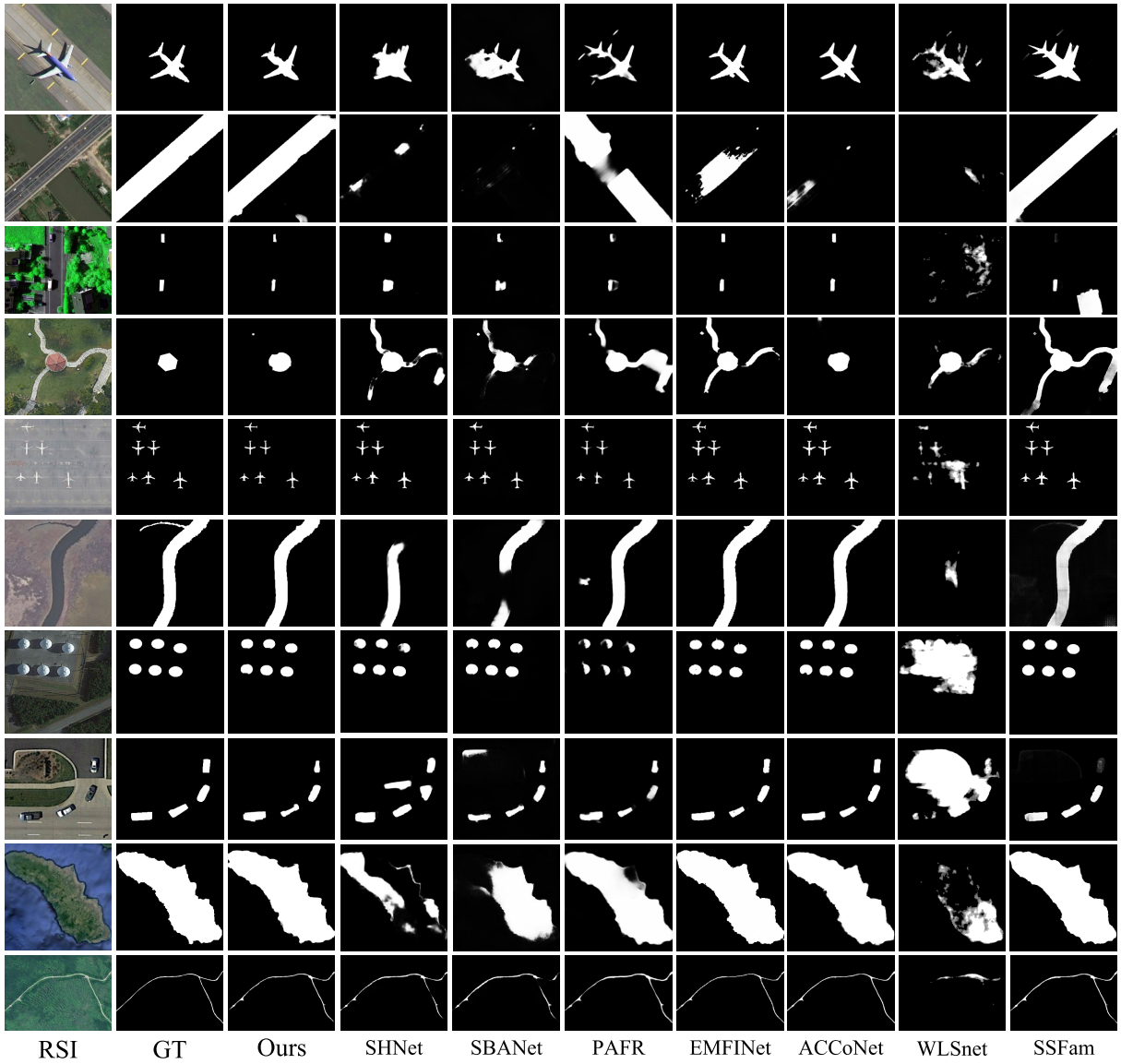


Fig. 6. Qualitative results.

TABLE II

ABLATION EXPERIMENTS FOR MODULE EFFECTIVENESS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

	Module					EORSSD				ORSSD			
	baseline	FRL	MEP	CESI	Params	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	✓				2.25M	0.8067	0.9401	0.8732	0.0118	0.8369	0.9374	0.8781	0.0191
b)	✓	✓			4.21M	0.8206	0.9495	0.8677	0.0103	0.8581	0.9581	0.8797	0.0168
c)	✓		✓		2.42M	0.8357	0.9549	0.8865	0.0100	0.8614	0.9594	0.8861	0.0161
d)	✓		✓	✓	2.85M	0.8435	0.9608	0.8928	0.0094	0.8669	0.9628	0.8919	0.0158
e)	✓	✓	✓	✓	4.81M	0.8475	0.9628	0.8871	0.0081	0.8846	0.9690	0.9007	0.0127

with $\text{DSConv}_{d=4}()$. It shows worse performance than d), which verifies that the Mamba branch can compensate for the limitations of the receptive field in traditional convolution operation.

Furthermore, to validate the generalization ability of FRL, we injected it into different ORSI-SOD models (CoorNet [45], SeaNet [46], and MEANet [49]) and retrained them with our training strategy. The comparative results are presented in Table IV. It clearly demonstrates that all four metrics are improved after incorporating the FRL, validating its capability to compensate for the inadequate feature representation of lightweight backbones. The proposed FRL is an effective plug-and-play component.

3) Effectiveness of the MEP Module: As shown in Fig. 3, we divide the MEP module into two stages and conduct experiments to validate each stage, a) w/o x_1 (removing

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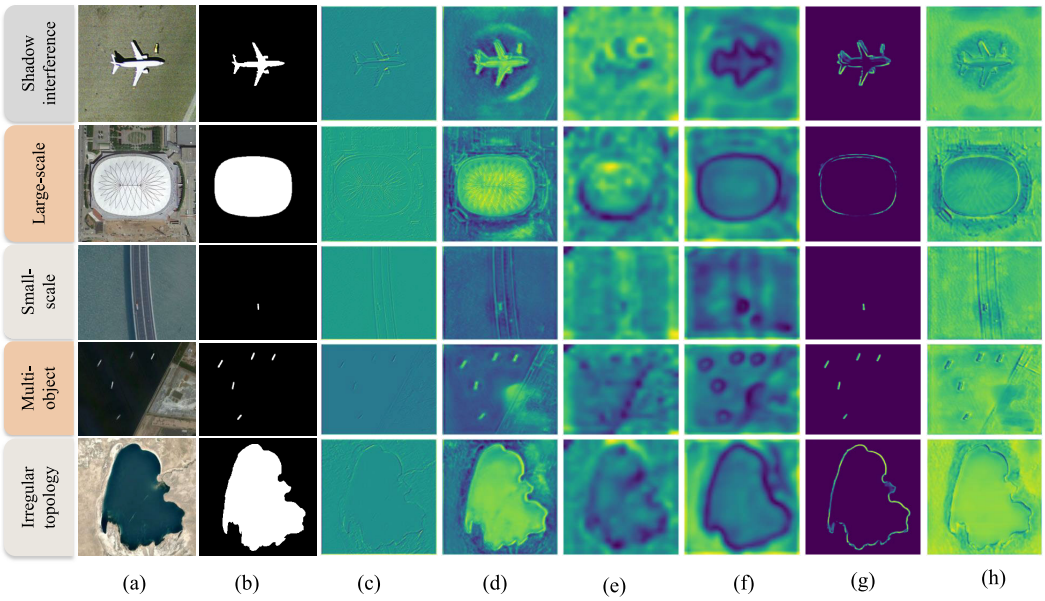


Fig. 7. Sample features for each module. (a) RSI, (b) GT, (c) E^1, (d) DAU_1, (e) E^4, (f) MFA_4, (g) MEP, and (h) CESI.

TABLE III
ABLATION EXPERIMENTS OF FRL. THE BEST RESULTS ARE HIGHLIGHTED IN RED

		EORSSD				ORSSD			
	Methods	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	all_DAU	0.8322	0.9559	0.8801	0.0094	0.8628	0.9580	0.8885	0.0149
b)	all_MFA	0.8413	0.9558	0.8901	0.0084	0.8730	0.9601	0.8952	0.0135
c)	w/o Mamba	0.8386	0.9572	0.8881	0.0083	0.8730	0.9634	0.8997	0.0137
d)	ours	0.8475	0.9628	0.8871	0.0081	0.8846	0.9690	0.9007	0.0127

TABLE IV
GENERALIZATION TEST OF FRL. * DENOTES THE METHOD ADDING FRL. THE BEST RESULTS ARE HIGHLIGHTED IN RED

	EORSSD				ORSSD			
Methods	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
CoorNet	0.8483	0.9529	0.9098	0.0122	0.8579	0.9453	0.9045	0.0215
CoorNet*	0.8736	0.9643	0.9263	0.0085	0.8934	0.9643	0.9264	0.0127
SeaNet	0.8457	0.9595	0.9081	0.0093	0.8764	0.9592	0.9141	0.0144
SeaNet*	0.8497	0.9628	0.9104	0.0087	0.8779	0.9608	0.9144	0.0139
MEANet	0.8770	0.9711	0.9309	0.0078	0.8850	0.9608	0.9249	0.0142
MEANet*	0.8888	0.9734	0.9364	0.0064	0.8988	0.9667	0.9332	0.0108

TABLE V
ABLATION EXPERIMENTS OF MEP. THE BEST RESULTS ARE HIGHLIGHTED IN RED

		EORSSD				ORSSD			
		F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	w/o x_1	0.8296	0.9551	0.8751	0.0103	0.8805	0.9685	0.8979	0.0135
b)	w/o x_3	0.8401	0.9575	0.8927	0.0086	0.8764	0.9619	0.8973	0.0138
c)	ours	0.8475	0.9628	0.8871	0.0081	0.8846	0.9690	0.9007	0.0127

shallow feature extraction), b) w/o x_3 (discarding high-level weighting), and c) ours. From Table V, we can see that compared to c), a) shows a noticeable increase in MAE on both datasets, indicating the effectiveness of our shallow feature processing strategy. Compared to c), the S_α metric of b) shows a slight increase on the EORSSD dataset, while other metrics show varying degrees of degradation. This indicates that our high-level weighting strategy achieves the best overall performance.

4) Effectiveness of the CESI Module: To verify its effectiveness, we conduct ablation experiments from two perspectives: removing key operations and replacing them with other modules. a) w/o lastmamba, b) Conv (replace Mamba with convolution), c) Attention (replace Mamba with self-attention mechanism), d) SS2D (replace Mamba with 2-D Selective Scan), and e) Ours. As shown in Fig. 4, considering the interaction between the two features, we eliminate the last Mamba block to assess its necessity. As shown in Table VI, we can see that although the parameters have slightly decreased compared to e), the performance has significantly declined. It indicates that our feature fusion strategy using the Mamba block can yield better results.

In the component replacement experiments, we replace the original blocks with Conv, Self-Attention, and SS2D in Table VI b)–d). The performance of b) and c) has a notable decline, and c) also requires expensive computational costs. Our analysis of SS2D in d) reveals that this mechanism not only significantly increases computational overhead but also leads to notable performance degradation. The possible reason is that with each additional scanning direction, redundancy increases and inference becomes slower. The results demonstrate that the Mamba-based fusion architecture not

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TABLE VI
ABLATION EXPERIMENTS OF CESI. THE BEST RESULTS ARE HIGHLIGHTED IN RED

				EORSSD				ORSSD			
		Params	GFLOPs	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	w/o lastMamba	4.64M	18.86	0.8391	0.9571	0.8746	0.0091	0.8670	0.9606	0.8830	0.0152
b)	Conv	9.19M	21.00	0.8262	0.9522	0.8950	0.0096	0.8549	0.9560	0.8910	0.0162
c)	Attention	5.63M	48.86	0.8267	0.9457	0.8870	0.0119	0.8588	0.9546	0.8993	0.0166
d)	SS2D	6.06M	108.48	0.7685	0.8994	0.8361	0.0462	0.7926	0.8975	0.8451	0.0370
e)	Ours	4.81M	23.03	0.8475	0.9628	0.8871	0.0081	0.8846	0.9690	0.9007	0.0127

TABLE VII
ABLATION EXPERIMENTS RESULTS FOR THE BACKBONE NETWORK. THE BEST RESULT IS HIGHLIGHTED IN RED

			EORSSD				ORSSD			
	Backbone	Params	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	VGG	120.7M	0.8488	0.9578	0.8961	0.0099	0.8739	0.9605	0.8919	0.0164
b)	SwinT	44.82 M	0.7753	0.9299	0.8596	0.0133	0.8204	0.9455	0.8677	0.0203
c)	PVT	33.32 M	0.8403	0.9632	0.8968	0.0073	0.8859	0.9721	0.9076	0.0117
d)	Ours	4.81M	0.8475	0.9628	0.8871	0.0081	0.8846	0.969	0.9007	0.0127

TABLE VIII
VALIDATION OF THE FUSION MANNERS. THE BEST RESULT IS HIGHLIGHTED IN RED

		EORSSD				ORSSD			
	Fusion manners	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$	F_β^{max}	E_ξ^{max}	S_α	$\mathcal{M}$
a)	E_{123}-E_{45}	0.8292	0.9525	0.8923	0.0098	0.8619	0.9560	0.8911	0.0169
b)	Ours	0.8475	0.9628	0.8871	0.0081	0.8846	0.9690	0.9007	0.0127

only maintains a lightweight nature but also achieves superior detection performance.

5) Ablation Experiment for the Backbone Network: To confirm the superiority of MobileNetV2 in our model, we perform comparative experiments with different backbones: a) Swin Transformer (four-layer backbone features will discard f_3), b) VGG, c) PVT, and d) Ours (MobileNetV2). In Table VII, the poor results of a) and b) indicate that the convolution operations of a) and the sliding window mechanism of b) both limit their multiscale perception capabilities. Rows c) and d) exhibit better generalization capabilities. Although c) shows superior performance, d) can significantly reduce parameters and computational burden through depth-wise separable convolutions, enabling it to be suitable for deployment on edge devices.

6) Effectiveness of the Interaction Manners: To verify the effect of interaction fusion strategy, we compared different interaction manners: a) E_{123}–E_{45} (use f_e^1, f_e^2, and f_e^3 for edge extraction, and f_e^4 and f_e^5 for fusion) and b) Ours. As shown in Table VIII, most metrics of a) exhibit a notable decrease. The results demonstrate that weighted fusion and high-level constraint strategy significantly improve the performance of the proposed model, proving the superiority of this strategy.

D. Discussion

Although the LightEMNet demonstrates high computational efficiency and low labeling cost, it has some limitations in certain complex scenarios. First, applying a weighted operation to reduce parameters leads to decreased edge continuity. As shown in row 2 in Fig. 7(g), the object's edge appears disconnected. Furthermore, the reduction of convolutional layers leads to insensitivity to details, as illustrated in rows 6–8 in Fig. 6, which shows its ineffectiveness for detail processing of small targets. In summary, our pursuit of lightweight design comes at the cost of detection accuracy, resulting in a performance gap compared to fully supervised methods. Despite these advancements, we identify several promising directions for future research: 1) generalization enhancement. Improving model adaptability to various remote sensing applications, including ship detection and road extraction tasks; 2) an adaptive edge extraction module. Future research should explore adaptive mechanisms to improve edge continuity in different scenes; and 3) performance gap reduction. Existing weakly supervised ORSI-SOD methods still have a performance gap with the latest fully supervised methods, so exploring higher-accuracy methods is also one of the important directions in the future.

V. CONCLUSION

In this article, we propose a lightweight weakly supervised framework for ORSI-SOD, which employs MobileNetV2 as the backbone for feature extraction. An FRL based on a hierarchical optimization strategy is designed to address the feature degradation caused by a lightweight backbone, which significantly enhances feature representation. Additionally, we consider the utilization of edge information from a unique perspective, emphasizing the importance of high-level information constraints in edge extraction. Therefore, we propose a multilayer edge perception (MEP) module for edge prediction, which not only refines the edges but also achieves accurate object localization. Furthermore, to fully leverage the edge features obtained from MEP and high-level semantics for calibrating objects, we propose a Mamba-based CESI module. The addition of Mamba blocks improves the

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model's adaptability to complex topological structures while preserving a lightweight architecture. Extensive experiments on public datasets demonstrate that our proposed LightEMNet achieves superior performance with fewer parameters and lower computational complexity.

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